

Geospatial Data Science: Project Report Template

Student 1 name & Student 2 name

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Abstract

We suggest to include an abstract in the beginning of your project report, max 200 words. Suggested abstract structure: 1-2 sentences basic introduction into field. 2-3 sentences more detailed background. 1 sentence clearly stating the general problem being addressed by this particular study. 1 sentence summarizing the main result (“here we show”). 2-3 sentences explaining what the main result reveals/adds. 1-2 sentences to put results into more general context. Annotated best practice example (from Nature), where you can leave out the last *broader perspective* part: <https://www.nature.com/documents/nature-summary-paragraph.pdf>

Technicalities

Layout

- font size: 11pt
- margins: 1.5 cm
- linespacing: single
- pages (without references and appendices):
 - 5-9 pages for groups of size 1
 - 7-11 pages for groups of size 2
 - 9-13 pages for groups of size 3
- must contain at least one figure with geospatial data visualization
- all tables and figures labelled
- Collaboration – e.g. on overleaf: <https://www.overleaf.com/>

Code repository

If your project is a prototype of a digital product, don’t hand in the code – instead, include in the report:

- a link to your code repository
- an URL (or another way) where your prototype can be tried out

If your project is based on a research question, hand in:

- gitlog.txt: Your repo's git log, e.g. by running: `git log > gitlog.txt`
- code.zip: One zip file containing Jupyter notebook(s) (.ipynb) of your commented code that runs fully without errors, assuming that all used data files are in place, and reproduces your findings. Do not include the data files here.

1 Introduction

Briefly motivate **why** your project outcomes are relevant. If you worked with a research question, state it here. If you created a prototype, state what the prototype is intending to do.

2 Background

Provide **context** for your project. What has happened so far? What problems does your project address, and what is the role of digital tools for solving these problems? Outline your approach.

3 Data

3.1 Data acquisition

Where does your data come from? Comment on their fitness for purpose, focusing on the geospatial aspect (format, quality, provenance, precision, reliability)

3.2 Data processing

Provide details of data manipulation and transformation. Name challenges, if any. Link to processing scripts where relevant.

4 Results

- State results here
- Illustrate with figures where relevant (unlike figure 1).
- Describe main workflow elements
- Name challenges and bottlenecks (but leave main discussion/critical evaluation to the next section)
- Functions, tips & tricks you want to promote or credit
- For 'technical pipeline' projects: provide a guided tour of your pipeline to facilitate its reproducibility, explaining your choices, clarifying dependencies, and referring to the scripts/tools you compiled in GitHub.



Figure 1: A completely unrelated frog.

5 Discussion

- Evaluate your results
- How representative, reliable, complete, precise, transferable, generalizable?
- What are the shortcomings of your approach?
- Name caveats, discarded approaches, technical difficulties, lessons learned the hard way
- If your project outcome is a digital prototype: compare with other state-of-the-art digital tools, if any, for the same task

6 Conclusion

- Summarize all of the above in a few sentences
- Achieved and/or missed goals
- Highlight lessons learned
- Future work needed: to address the shortcomings of your project; to further investigate in this field; to further improve/extend prototype functionalities

7 References

Choose whichever reference system you like - and stick to it. Cite some authors like Greenwade (1993) - again, you're free to pick any format as long as it's coherent. Examples:

- Chicago Style
- APA
- Harvard

References

Greenwade George D. The Comprehensive Tex Archive Network (CTAN) // TUGBoat. 1993. 14, 3. 342–351.

Appendix A: Meta data tables

Metadata tables – see project report for details

Appendix B: Contribution statement

For individualized grading, here you must clarify who of your group is responsible for the different parts of the project submission. Please state:

- For **at least 2 sections** of your report, who was **primary contributor** (and optionally also second and/or third contributor)
- For any **additional major tasks** such as data collection, data preparation, analysis, programming, or visualization, who was **primary contributor** (optionally also second and/or third contributor)